Versorgung

Versorgung: 1S-Pfad (Elektronik) und 2S-Pfad (Aktoren, Pyro)

Datei: power.kicad_sch

Teensy

Teensy 4.1 auf Buchsenleisten, Erweiterungsleiste, Befestigung

Datei: mcu.kicad_sch

Sensorik

IMU ICM-42688-P an SPI1, Barometer BMP390 und Qwiic an I2C

Datei: sensors.kicad_sch

Funk

LoRa E22-400T22S an Serial8, GPS an Serial1

Datei: radio.kicad_sch

Aktoren

TVC-Servos am 2S-Akku, Fallschirm-Servo, Rollmotor DRV8833

Datei: actuators.kicad_sch

Pyro

Zwei Pyro-Kanaele low-side, Arming, Durchgangspruefung

Datei: pyro.kicad_sch

Signal

WS2812B mit Pegelwandler, Summer, Kamera-Trigger

Datei: signal.kicad_sch

Mission Space – Flugcomputer-Mainboard rev B (54 x 120 mm, 4 Lagen, JLCPCB)

Spezifikation: hardware/mainboard/README.md; Design Guide: docs/DesignGuide_Mainboard_revB.md

Massen: GND = Logik und 1S-Minus; GND_PWR = Rueckstrom aller 2S-Verbraucher. Einzige Verbindung: NT1 am 2S-Minus.

Dieser Schaltplan wird von gen_schematic.py aus circuit.py erzeugt. Aenderungen dort machen, dann neu erzeugen.

Teensy 4.1, ICM-42688-P, BMP390, E22-400T22S, 1S + 2S

Mission Space

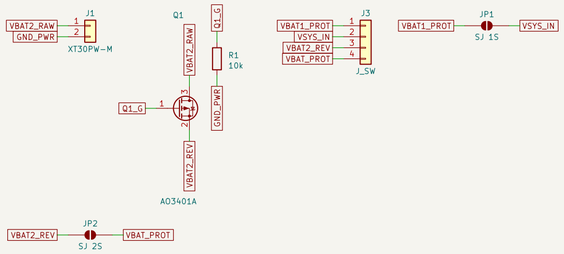
Sheet: /
File: mainboard.kicad_sch

Title: Mainboard rev B

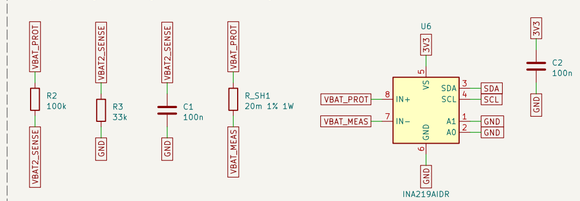
Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 1/8

Versorgung: 1S-Pfad (Elektronik) und 2S-Pfad (Aktoren, Pyro)

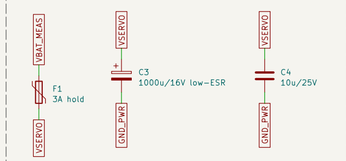
2S: Akku, Verpolschutz, Hauptschalter



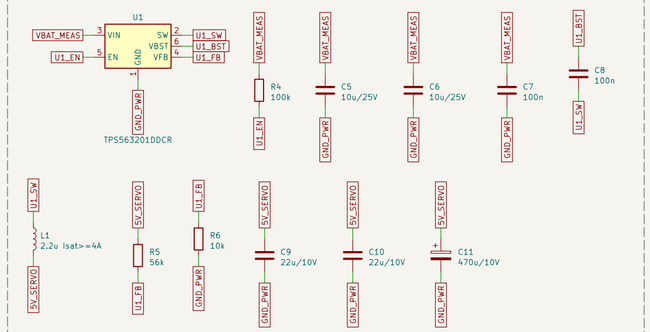
2S: Spannungsmessung und Strommessung INA219



2S: Servo-Schiene VSERVO



2S: Buck U1 -> 5V_SERVO

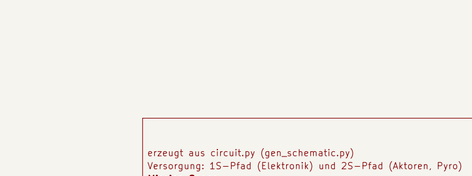


1S: Akku, Verpolschutz, Spannungsmessung

1S: Buck-Boost U2 -> 5V_SYS, Teensy-VIN

LDO U3 -> 3V3_RF, Sternpunkt, Testpunkte

Power-Flags fuer den ERC



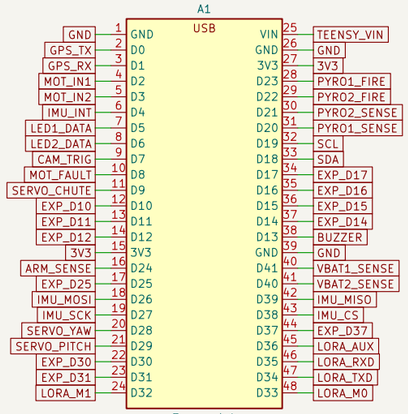
erzeugt aus circuit.py (gen_schematic.py)
Versorgung: 1S-Pfad (Elektronik) und 2S-Pfad (Aktoren, Pyro)
Mission Space
Sheet: /Versorgung/
File: power.kicad_sch

Title: Mainboard rev B – Versorgung

Size: A2	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 2/8

Teensy 4.1 auf Buchsenleisten, Erweiterungsleiste, Befestigung

Teensy 4.1



Erweiterung, Befestigung, Testpunkte



erzeugt aus circuit.py (gen_schematic.py)
Teensy 4.1 auf Buchsenleisten, Erweiterungsleiste, Befestigung
Mission Space

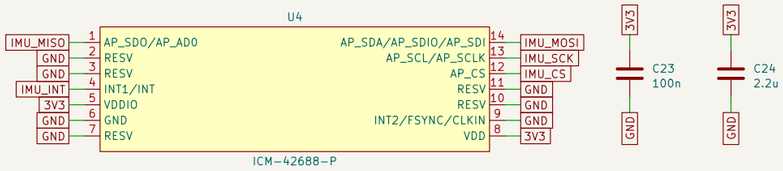
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File: mcu.kicad_sch

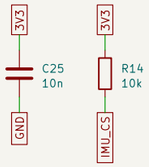
Title: Mainboard rev B – Teensy

Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 3/8

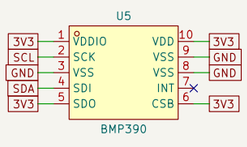
IMU ICM-42688-P an SPI1, Barometer BMP390 und Qwiic an I2C

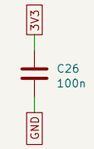
IMU ICM-42688-P, SPI1

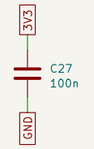


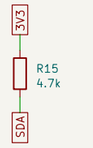


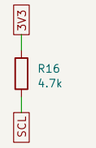
Barometer BMP390 und I2C

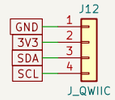












erzeugt aus circuit.py (gen_schematic.py)
IMU ICM-42688-P an SPI1, Barometer BMP390 und Qwiic an I2C

Mission Space

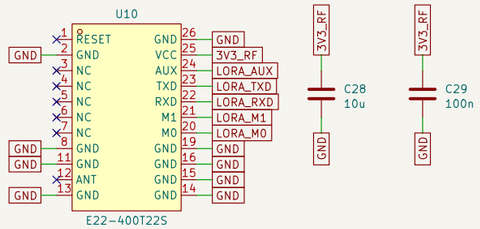
Sheet: /Sensorik/
File: sensors.kicad_sch

Title: Mainboard rev B – Sensorik

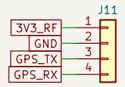
Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 4/8

LoRa E22-400T22S an Serial8, GPS an Serial1

LoRa E22-400T22S, Serial8



GPS, Serial1



erzeugt aus circuit.py (gen_schematic.py)
LoRa E22-400T22S an Serial8, GPS an Serial1
Mission Space

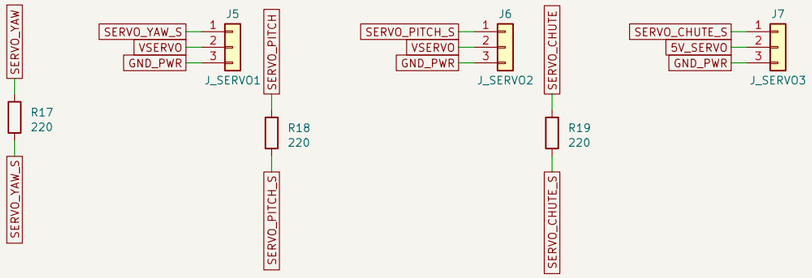
Sheet: /Funk/
File: radio.kicad_sch

Title: Mainboard rev B – Funk

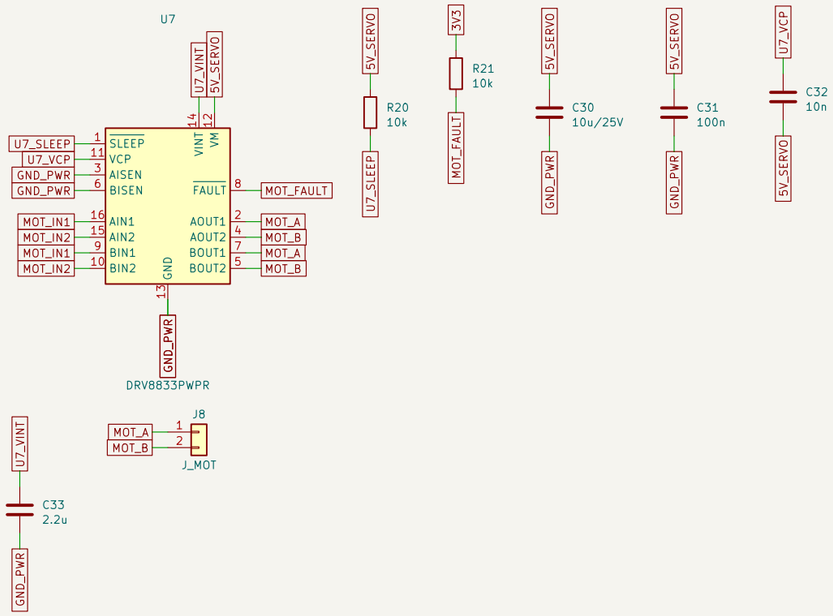
Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 5/8

TVC–Servos am 2S–Akku, Fallschirm–Servo, Rollmotor DRV8833

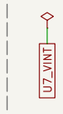
Servos



Rollmotor DRV8833



Power–Flags fuer den ERC



erzeugt aus circuit.py (gen_schematic.py)
TVC–Servos am 2S–Akku, Fallschirm–Servo, Rollmotor DRV8833
Mission Space

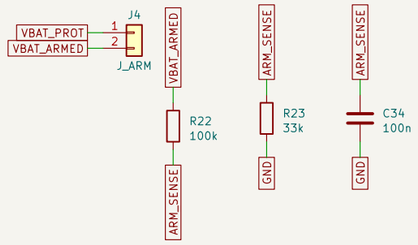
Sheet: /Aktoren/
File: actuators.kicad_sch

Title: Mainboard rev B – Aktoren

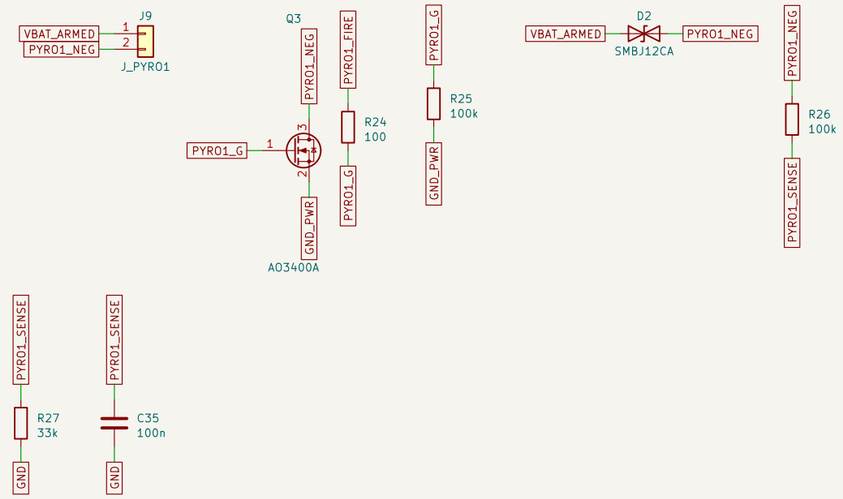
Size: A3	Date: 2026–10–21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 6/8

Zwei Pyro-Kanaele low-side, Arming, Durchgangspruefung

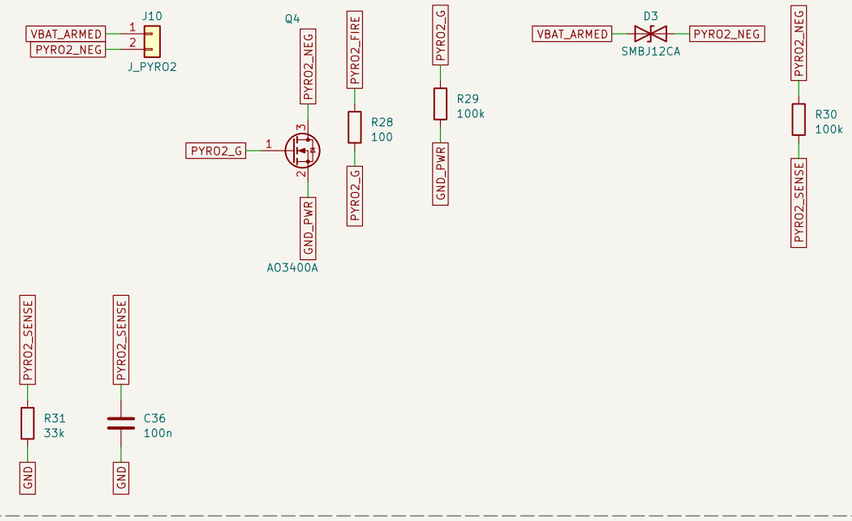
Arming



Pyro-Kanal 1



Pyro-Kanal 2



erzeugt aus circuit.py (gen_schematic.py)

Zwei Pyro-Kanaele low-side, Arming, Durchgangspruefung

Mission Space

Sheet: /Pyro/

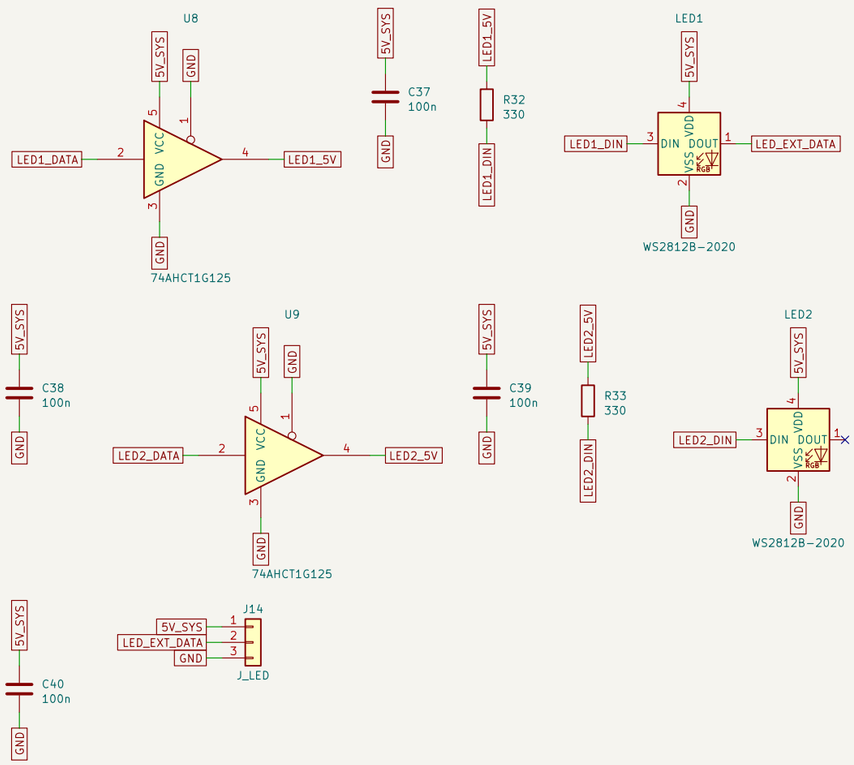
File: pyro.kicad_sch

Title: Mainboard rev B – Pyro

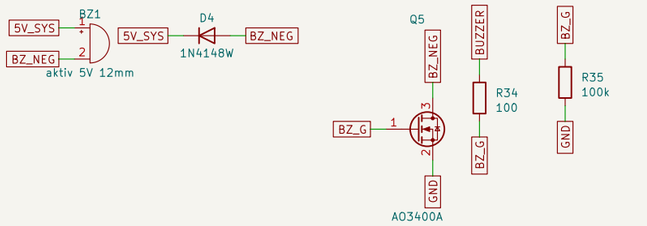
Size: A3	Date: 2026-10-21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 7/8

WS2812B mit Pegelwandler, Summer, Kamera–Trigger

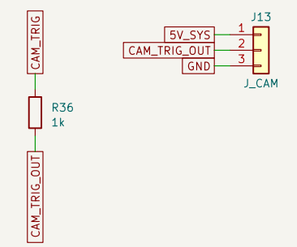
Status–LEDs



Summer



Kamera



erzeugt aus circuit.py (gen_schematic.py)
WS2812B mit Pegelwandler, Summer, Kamera–Trigger
Mission Space

Sheet: /Signal/		
File: signal.kicad_sch		
Title: Mainboard rev B – Signal		
Size: A3	Date: 2026–10–21	**Rev: B**
KiCad E.D.A. 10.0.6		Id: 8/8